Vibrational spectroscopy

The energy of a vibrating diatomic molecule:

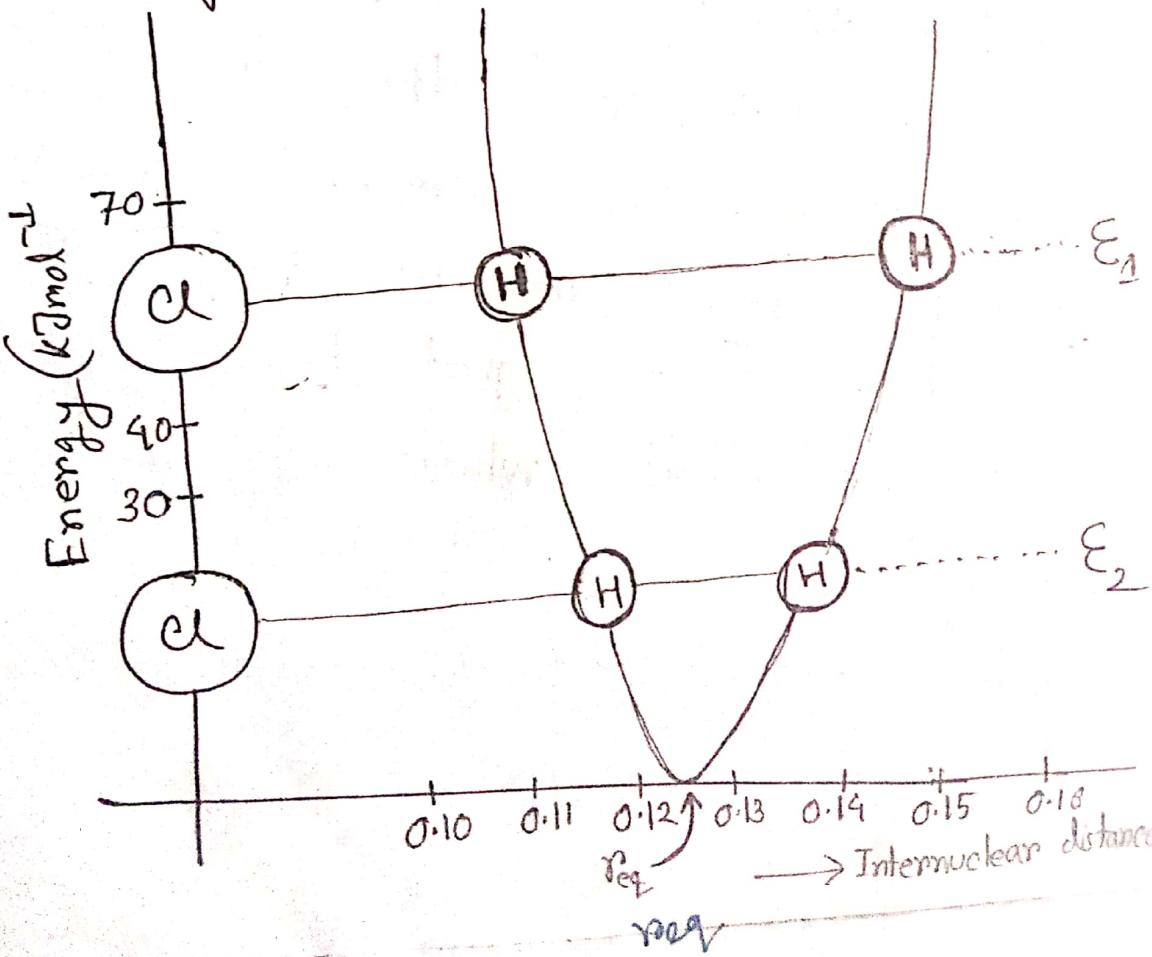
When two atoms combine to form a stable covalent molecule (e.g. HCl), they may be said to be formed by balancing two opposite forces.

There is a repulsion between the positively charged nuclei of both atoms and between the negative electron clouds. On the other hand there is an attraction between the nucleus of one atom and the electrons of the other and vice versa. The two atoms are settle at a mean distance such that these two forces just balance and the energy of the whole system is at a minimum.

Attempt to squeeze (সংকুচিত করা) the atoms more closely together and the repulsive force rises rapidly,

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attempt to pull them further apart and this is restricted by attractive force. In either case an attempt to distort the bond requires an input of energy. Following figure shows the energy against internuclear distance. The internuclear distance at minimum energy is called the equilibrium distance r_{eq} or the bond length.



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Fig: The energy of HCl as the bond is compressed or extended.

The compression and extension of a bond may be considered as like as a vibrating spring. So according to Hook's law —

$$f = -k(r - r_{eq}) \quad \ldots\ldots (i)$$

where, f is the restoring force, k is the force constant and r is the internuclear distance. The energy curve is parabolic and follows the eq^{n}s as written below—

$$E = \frac{1}{2} k (r - r_{eq})^2 \quad \ldots\ldots (ii)$$

This model of a diatomic molecule is so-called simple harmonic oscillator model.

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(10)

☐ Write down the energy equation for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule). ①

Soln: The energy expression for the harmonic oscillator is,

$$\mathcal{E}_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} \quad cm^{-1}$$

Where, $\bar{\omega}_{osc}$ = The oscillation frequency in cm^{-1}

~~v =~~ Rotational quantum number.

☐ Write down the energy expression for the anharmonic oscillator?

Soln:

The energy expression for the anharmonic oscillator is,

$$\mathcal{E}_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} - \left(v + \frac{1}{2}\right)^2 \bar{\omega}_{osc} \bar{x}_e \quad cm^{-1}$$

Where, $\bar{\omega}_{osc}$ = The oscillation frequency in cm^{-1}

$\bar{x}_e$ = Anharmonicity constant.

v = Vibrational quantum number.

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Write down the energy equation for a vibrating diatomic molecule. ① ②

Soln:

The combined rotational-vibrational energy expression is the sum of the separate energy -

$$E_{total} = E_{rot} + E_{vib} \quad \text{(Joules)}$$

$$\text{or, } \mathcal{E}_{total} = \mathcal{E}_{rot} + \mathcal{E}_{vib} \quad (cm^{-1}) \qquad \text{(i)}$$

Again, we know,

$$\mathcal{E}_{rot} = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \ldots\ldots \quad cm^{-1} \qquad \text{(ii)}$$

$$\mathcal{E}_{vib} = \left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e \quad cm^{-1} \qquad \text{(iii)}$$

From eqn (i), (ii) and (iii) we get -

$$\mathcal{E}_{total} = \mathcal{E}_{J,v} = \{BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \ldots\ldots\} + \left\{\left(v+\tfrac{1}{2}\right)\bar{\omega}_e - \left(v+\tfrac{1}{2}\right)^2 \bar{\omega}_e x_e\right\} \qquad \text{(iv)}$$

Initially we shall ignore the small centrifugal distortion constants D and H etc. Then we get -

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$$E_{total} = \left\{ B J(J+1) + \left(v+\frac{1}{2}\right)\bar{\omega}_e - \left(v+\frac{1}{2}\right)^2 \bar{\omega}_e x_e \right\} \text{ cm}^{-1}$$

Write down the selection rule for harmonic oscillator undergoing vibrational changes. (0.75)

Soln:
The selection rule for harmonic oscillator is,

$$\Delta v = \pm 1$$

Write down the selection rule for an anharmonic oscillator.

Soln:
The selection rule for anharmonic oscillator is, $\Delta v = \pm 1, \pm 2, \pm 3 \ldots\ldots$

Write down the selection rule for the diatomic vibrating rotator.

Soln: The selection rule for the diatomic vibrating rotator,

$\Delta v = \pm 1, \pm 2, \ldots$ etc ; $\Delta J = \pm 1$

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Show that a diatomic molecule which behave as a simple harmonic oscillator can never have zero vibrational energy.

Soln:

③ ①

An elastic bond, like a spring, has a certain intrinsic vibrational frequency, dependent on the mass of the system and the force constant, but is independent of the amount of distortion.

The oscillation frequency is –

$$\omega_{osc} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} \text{ Hz} \quad \text{——— (i)}$$

Where, μ is the reduced mass of the system.

To convert the frequency to wave-numbers –

$$\frac{\omega_{osc}}{c} = \bar{\omega}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \text{ cm}^{-1} \quad \text{——— (ii)}$$

Here, c = Velocity of light in cms^{-1}.

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The allowed vibrational energies for any particular system may be calculated from the schrodinger equation. For the simple harmonic oscillator these turn out to be:

$$E_v = \left(v + \frac{1}{2}\right) h\omega_{osc} \text{ joules} \quad (v = 0, 1, 2 \ldots\ldots) \qquad \text{(iii)}$$

Where, v is the vibrational quantum number.

Now, converting to the spectroscopic units cm^{-1} we have —

$$\varepsilon_v = \frac{E_v}{hc} = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} \ cm^{-1} \qquad \text{(iv)}$$

$$\left[\because \frac{\omega_{osc}}{c} = \bar{\omega}_{osc}\right]$$

energy expression

The lowest vibrational energy, obtained by putting $v = 0$ in eqn (iii) and (iv),

$$E_0 = \frac{1}{2} h\omega_{osc} \text{ joules} \quad (\omega_{osc} \text{ in Hz})$$

$$\varepsilon_0 = \frac{1}{2} \bar{\omega}_{osc} \ cm^{-1} \quad (\bar{\omega}_{osc} \text{ in } cm^{-1})$$

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So we can say that, the diatomic molecule can never have zero vibrational energy.

✱ Explain why transitions between any two neighbouring vibrational levels of a diatomic molecule give rise to the same energy.

Or, Show that the difference between two energy level is always equally spaced for simple harmonic oscillator. [আংক নি] (2)

Soln:

The energy expression of a diatomic molecule for a simple harmonic oscillator—

$$E_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{ose} \quad cm^{-1} \quad \ldots\ldots (i)$$

The simple selection rule for the harmonic oscillator undergoing vibrational changes—

$$\Delta v = \pm 1$$

Applying the selection rule for the transition betn neighbouring energy level, for emission—

$$\Delta\varepsilon = \varepsilon_{v+1} - \varepsilon_v = \left(v+1+\frac{1}{2}\right)\bar{\omega}_{osc} - \left(v+\frac{1}{2}\right)\bar{\omega}_{osc}$$

$$= v\bar{\omega}_{osc} + \bar{\omega}_{osc} + \frac{1}{2}\bar{\omega}_{osc} - v\bar{\omega}_{osc} - \frac{1}{2}\bar{\omega}_{osc}$$

$$= \bar{\omega}_{osc} \ cm^{-1}$$

Similarly, for absorption,

$$\Delta\varepsilon = \varepsilon_{v \to v+1} = \bar{\omega}_{osc} \ cm^{-1}$$

[Whatever the initial value of v]

It is seen that, the energy difference betn only two neighbouring vibrational energy level is same which indicates that all vibrational energy levels are equally spaced. Since vibrational energy levels are equally spaced, transitions betn any two neighbouring states will rise to the same energy change.

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Explain the lines are equally spaced in simple harmonic oscillator, so energy equal, whereas, in anharmonic oscillator energy level decreased with v risen. [আসবে নি]

Soln:

The vibrational energy for simple harmonic oscillator and and anharmonic oscillator respectively,

$$\varepsilon_v = \left(v + \frac{1}{2}\right) \bar{\omega}_e \quad \ldots\ldots (i)$$

$$\varepsilon_v = \bar{\omega}_e \left\{1 - x_e\left(v + \frac{1}{2}\right)\right\}\left(v + \frac{1}{2}\right) \quad \ldots\ldots (ii)$$

Here we can see from energy expression that in harmonic oscillator, energy rises with increasing v value. For the transition betn neighbouring energy, ... (আসবে question খুব কম করবে)

But in anharmonic oscillator, with increasing v value total energy decreased. That's why, in harmonic oscillator the lines of

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vibrational energy levels are equal spaced and in anharmonic oscillator the gap lines are decreased in vibrational energy level with increasing ν and they are not equally spaced.

☐ Sketch the Morse curve for the anharmonic oscillator. Explain fundamental, hot band and overtone transitions with energy expressions. (1+3)

Or, Draw the morse curve for the anharmonic oscillator. Explain fundamental, hot band and overtone transitions showing them in the curve, Write the energy expressions for the fundamental, hot band and overtone transitions. (5)

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The Morse curve for the anharmonic oscillator is given below—

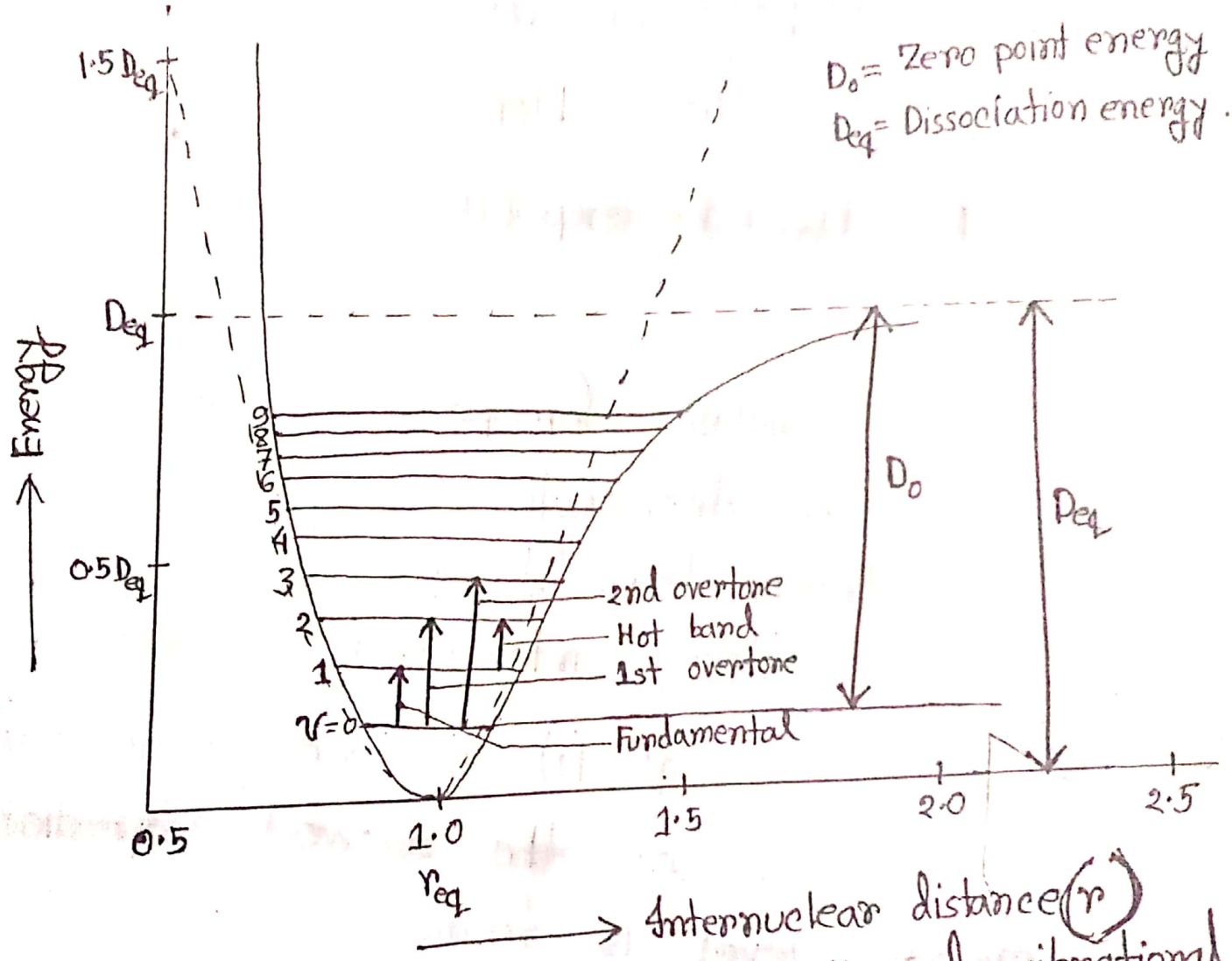


Fig: The morse curve; the allowed vibrational energy levels and some transitions between them for a diatomic molecule undergoing anharmonic oscillations.

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The energy expression of a diatomic molecule undergoing anharmonic extension and compression, was derived by Morse and is called Morse function -

$$E = D_{eq}\left[1 - \exp\{a(r_{eq} - r)\}\right]^2 \quad \ldots\ldots (1)$$

Where,

a = A constant for particular molecule.

D_{eq} = The dissociation energy.

r = Internuclear distance

r_{eq} = Equilibrium internuclear distance

Now using eqn (i) in schrodinger equation, the pattern of the allowed vibrational energy level is found to be -

$$\varepsilon_v = \left(v + \frac{1}{2}\right)\bar{\omega}_e - \left(v + \frac{1}{2}\right)^2 \bar{\omega}_e x_e \ cm^{-1}$$

Where,

$\bar{\omega}_e$ = The oscillation frequency in cm^{-1}

x_e = Anharmonicity constant.

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(a) Fundamental absorption:

Fundamental vibrational frequencies of a molecule corresponds to transition from $v=0$ to $v=1$ with considerable intensity.

$$\Delta\varepsilon = \varepsilon_{v=1} - \varepsilon_{v=0}$$

$$= \left(1+\frac{1}{2}\right)\bar{\omega}_e - \left(1+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= \frac{3}{2}\bar{\omega}_e - \frac{9}{4}\bar{\omega}_e x_e - \frac{1}{2}\bar{\omega}_e + \frac{1}{4}\bar{\omega}_e x_e$$

$$= \frac{1}{4}\bar{\omega}_e\left(6 - 9x_e - 2 + x_e\right)$$

$$= \frac{1}{4}\bar{\omega}_e\left(4 - 8x_e\right)$$

$$= \frac{1}{4}\bar{\omega}_e \cdot 4\left(1 - 2x_e\right)$$

$$= \bar{\omega}_e\left(1 - 2x_e\right) \ cm^{-1}$$

$$\therefore \Delta\varepsilon = \bar{\omega}_e\left(1 - 2x_e\right) \ cm^{-1}$$

* What is fundamental band? How much energy is involved in the fundamental transitions for the harmonic oscillator model for a diatomic molecule with fundamental frequency ω_e? (2)

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(b) Overtone band:

An overtone band is the spectral band that occur in vibrational spectrum of a molecule when the molecule makes a transition from the ground state ($v=0$) to the second or third excited state ($v=2$ or $v=3$)

For 1st overtone:

$v=0 \rightarrow v=2$, $\Delta v = +2$ with small intensity.

$$\Delta\varepsilon = \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right\}$$

$$= \frac{5}{2}\bar{\omega}_e - \frac{25}{4}\bar{\omega}_e x_e - \frac{1}{2}\bar{\omega}_e + \frac{1}{4}\bar{\omega}_e x_e$$

$$= \frac{1}{4}\bar{\omega}_e\left(10 - 25x_e - 2 + x_e\right)$$

$$= \frac{1}{4}\bar{\omega}_e\left(8 - 24x_e\right)$$

$$= \frac{1}{4}\bar{\omega}_e \cdot 8\left(1 - 3x_e\right)$$

$$= 2\bar{\omega}_e\left(1 - 3x_e\right) \ cm^{-1}$$

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For 2nd overtone:

$v=0 \longrightarrow v=3$, $\Delta v=+3$ with normally negligible intensity.

$\therefore \quad \Delta\varepsilon = 3\bar{\omega}_e(1-4x_e) \quad cm^{-1}$

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(c) Hot bands:

The weak absorption arise at slightly lower wave number than the fundamental. Such weak absorption is known as hot bands. A high temperature is the preliminary condition for their appearance.

Here $v=1 \longrightarrow v=2$, $\Delta v=+1$

Transition energy, $\Delta\varepsilon = \varepsilon_{v=2} - \varepsilon_{v=1}$

$$= \left(2+\frac{1}{2}\right)\bar{\omega}_e - \left(2+\frac{1}{2}\right)^2 \bar{\omega}_e x_e - \left\{\left(1+\frac{1}{2}\right)\bar{\omega}_e - \left(1+\frac{1}{2}\right)^2 \bar{\omega}_e x_e\right\}$$

$$= \bar{\omega}_e(1-4x_e)$$

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What are the different possible transitions for a diatomic molecule undergoing anharmonic oscillations? Describe the transitions with energy expressions, energy diagram and spectrum. (1+4)

Soln:

The different possible transitions for a diatomic molecule undergoing anharmonic oscillation:

(1) Fundamental absorption:

$v=0 \rightarrow v=1$, $\Delta v=+1$, with considerable intensity.

(2) First overtones:

$v=0 \rightarrow v=2$, $\Delta v=+2$ with small intensity.

Second overtones:

$-0 \rightarrow v=3$, $\Delta v=+3$ with normally

eligible intensity.

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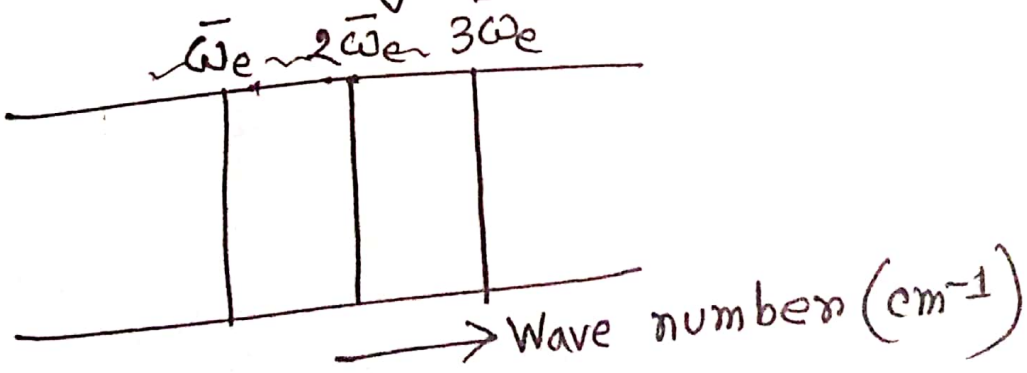
(4) Hot bands :

$v=1 \longrightarrow v=2$, $\Delta v = +1$, normally very weak.

The Morse curve এর figure

আগের question এর Morse curve থেকে খুব্বু করে একদম শেষ পর্যন্ত , তারপর নিচেরটুকু +

The spectrum arising from them —



Spectrum.

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Explain why the spacing between vibrational levels in a Morse curve decreases gradually? (2)

Soln:

In anharmonic molecule, the average bond length increases with increasing the vibrational energy.

$$B = \frac{h}{8\pi^2 I c}$$

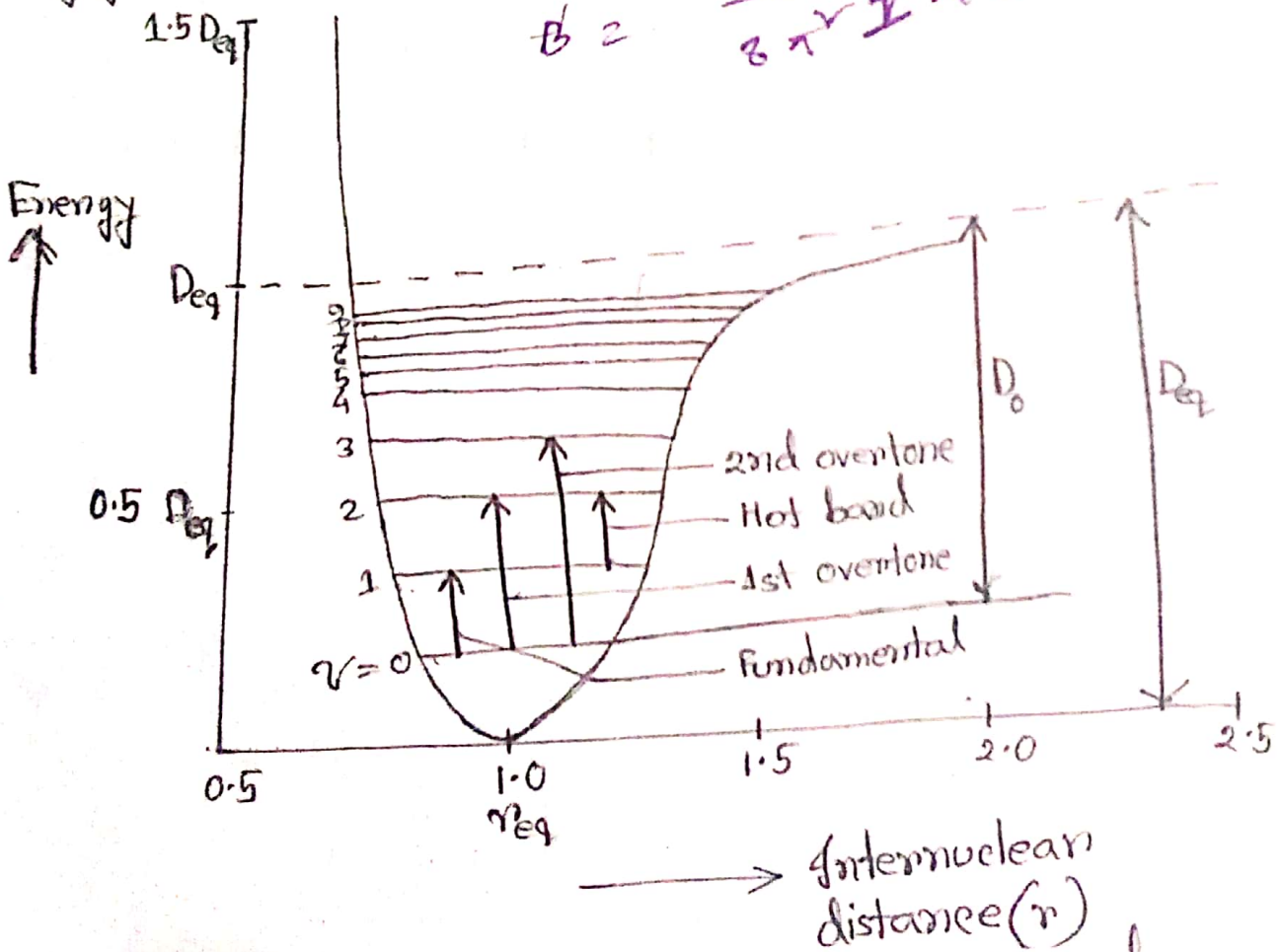


Fig: The morse curve; the allowed vibrational energy levels and some transitions betn them for a diatomic molecule undergoing anharmonic oscillation.

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The rotational constant (B) is inversely proportional to the average bond length. So, the spacing between vibrational levels in a Morse curve decreases gradually.

What are the zero-point energies in relation to harmonic and anharmonic oscillators? ① ②

Soln:

The allowed energy level to a simple harmonic oscillator is —

$$\varepsilon_v = \left(v + \frac{1}{2}\right) \bar{\omega}_{osc} \ cm^{-1} \quad \ldots\ldots (i)$$

The zero point vibrational energy ($v=0$) for harmonic oscillator —

$$\varepsilon_0 = \left(0 + \frac{1}{2}\right) \bar{\omega}_{osc}$$

$$\therefore \varepsilon_0 = \frac{1}{2} \bar{\omega}_{osc} \quad \ldots\ldots (ii)$$

The allowed energy level expression for anharmonic oscillator —

$$\varepsilon_v = \left(v + \frac{1}{2}\right) \bar{\omega}_e - \left(v + \frac{1}{2}\right)^2 \bar{\omega}_e x_e \ cm^{-1} \quad \ldots\ldots (iii)$$

where, $v = 0, 1, 2, 3 \ldots$

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The zero point vibrational energy ($v=0$) for the anharmonic oscillator–

$$\varepsilon_0 = \frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e \quad \text{.....(iv)}$$

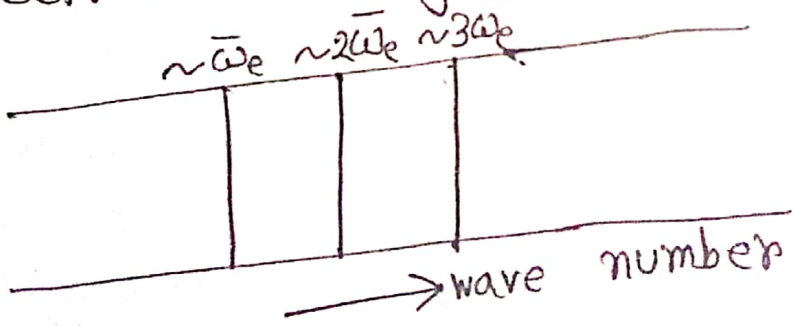
Therefore the zero point energy for anharmonic oscillator is slightly different from the harmonic oscillator.

Sketch a schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic oscillator. Show the allowed transitions and the expected spectrum? (4)

Soln:

The Morse curve

The spectrum arising from them,



Spectrum.

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The fundamental frequency of vibration for $^1H^{35}Cl$ molecule is 2886 cm^{-1}. Calculate the force constant of the $^1H^{35}Cl$ molecule. Given that mass of $^{35}Cl = 34.97$ amu. (3)

Soln:

We know,

force constant $k = 4\pi^2 \bar{\omega}_e^2 c^2 \mu$

Here, $\bar{\omega}_e = 2886\ cm^{-1}$

$= 2886 \times 100\ m^{-1}$

$c = 3 \times 10^8\ ms^{-1}$

$\mu = \frac{m_1 \times m_2}{m_1 + m_2}$

$= \left(\frac{1.0078 \times 34.97}{1.0078 + 34.97}\right) \times 1.66 \times 10^{-27}$

$= 1.626 \times 10^{-27}$

So, $k = 4\pi^2 \bar{\omega}_e^2 c^2 \mu$

$= 4 \times (3.1416)^2 \times (288600)^2 \times (3 \times 10^8)^2 \times 1.626 \times 10^{-27}$

$= 481.214\ Nm^{-1}$ (Ans)

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The equilibrium oscillation frequency for NO is 1904 cm^{-1}. Calculate the force constant of the molecule. (2.5)

Soln:

We know that,

$$K = 4\pi^2 \bar{\omega}_e c^2 \mu$$

Given that,

$$\bar{\omega}_e = 1904\ cm^{-1}$$

$$= (1904 \times 100)\ m^{-1}$$

$$\mu = \left(\frac{14 \times 16}{14+16}\right) \times 1.66 \times 10^{-27}$$

$$= 1.24 \times 10^{-26}\ kg$$

$$c = 3 \times 10^8\ ms^{-1}$$

$$\therefore K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu$$

$$= 4 \times (3.1416)^2 \times (190400)^2 \times (3 \times 10^8)^2 \times 1.24 \times 10^{-26}$$

$$= 1597.19\ Nm^{-1}$$

Ans

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The fundamental band of CO in inert CCl_4 solvent is found at 2143 cm^{-1} when examined in the IR region. Calculate the force constant of the molecule. (3)

Soln:

$$\mu = \left(\frac{12 \times 16}{12+16}\right) \times 1.66 \times 10^{-27}$$

$$= 1.14 \times 10^{-27}$$

We know,

$$K = 4\pi^2 \bar{\omega}_e^2 c^2 \mu^2$$

$$= 4 \times (3.14)^2 \times (214300)^2 \times (3 \times 10^8)^2 \times 1.14 \times 10^{-26}$$

$$= 1860.16 \ Nm^{-1}$$

The fundamental mode of HCl occurs at 2886 cm^{-1}. Predict the frequency of the corresponding mode of DCl. (1)

Soln:

Let k is equal for H–Cl and D–Cl

$$\therefore \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \frac{\sqrt{\frac{K_1}{\mu_1}}}{\sqrt{\frac{K_2}{\mu_2}}}$$

$$\Rightarrow \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \sqrt{\frac{\mu_2}{\mu_1}} \quad \text{..... (i)}$$

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$$\therefore \quad \mu_1 = \frac{1 \times 35}{1+35}$$

$$= \frac{35}{36}$$

$$\therefore \quad \mu_2 = \frac{2 \times 35}{2+35}$$

$$= \frac{70}{37}$$

$$\therefore \quad \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = \sqrt{\frac{70/37}{35/36}} = \sqrt{\frac{70 \times 36}{37 \times 35}}$$

$$\Rightarrow \quad \frac{\nu_{H-Cl}}{\nu_{D-Cl}} = 1.39$$

$$\Rightarrow \quad \nu_{D-Cl} = \frac{\nu_{H-Cl}}{1.39}$$

$$= \frac{2886}{1.39}$$

$$= 2068 \ cm^{-1} \quad (Ans)$$

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[7] Assume the following data for the molecule $^1H^{35}Cl$:

Bond length : 127.5 pm (ℓ), Bond force constant : 516.3 Nm^{-1} (K)

Atomic masses : $^1H = 1.673 \times 10^{-27}$ kg

$^{35}Cl = 58.06 \times 10^{-27}$ kg

Calculate the zero point energy, the energy of the fundamental vibration ν_0 and the rotational constant B.

Soln:

Given, force constant, $k = 516.3\ Nm^{-1}$ ✓

For $^1H^{35}Cl$, reduced mass, $\mu = \frac{1 \times 35}{1+35} \times 1.675 \times 10^{-27}$ kg ✓

$= 1.63 \times 10^{-27}$ kg ✓

∴ The oscillation frequency is,

$$\omega_{osc} = \frac{1}{2\pi} \times \sqrt{\frac{k}{\mu}}$$

$$= \frac{1}{2\pi} \sqrt{\frac{516.3}{1.63 \times 10^{-27}}}$$

$= 8.95 \times 10^{13}$ Hz

Rough: [$c = 3 \times 10^8$ m $= 3 \times 10^{10}$ cm]

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We know, Vibrational energy, $E_v = \left(v + \frac{1}{2}\right)\bar{\omega}_{osc}$

$\rightarrow E_v = \left(v+\frac{1}{2}\right)h\omega_{osc}$

*Now, the zero point energy is,

$$E_{v=0} = \frac{1}{2} h \omega_{osc} \text{ Joules}$$

$$= \frac{1}{2} \times 6.63 \times 10^{-34} \times 8.95 \times 10^{13} \text{ J}$$

$$= 2.97 \times 10^{-20} \text{ J} \quad \text{(Ans)}$$

Again, The energy of fundamental vibration ν_0 is $E_0 = \frac{1}{2} h \omega_{osc}$ J

$= 2.97 \times 10^{-20}$ J (Ans)

$E_v = \left(v+\frac{1}{2}\right) h \omega_{osc}$ J

And, We know, $I = \mu r^2$

$= 1.63 \times 10^{-27} \times 127.5 \times 10^{-12}$

$= 2.1 \times 10^{-37}$ kg m^2

Given,
$r = 127.5$ pm
$= 127.5 \times 10^{-12}$ m

$$\therefore B = \frac{h}{8\pi^2 I C}$$

$$= \frac{6.63 \times 10^{-34}}{8\pi^2 \times 2.1 \times 10^{-37} \times 3 \times 10^8}$$

$$= 6.5 \times 10^{27} \text{ m}^{-1}$$

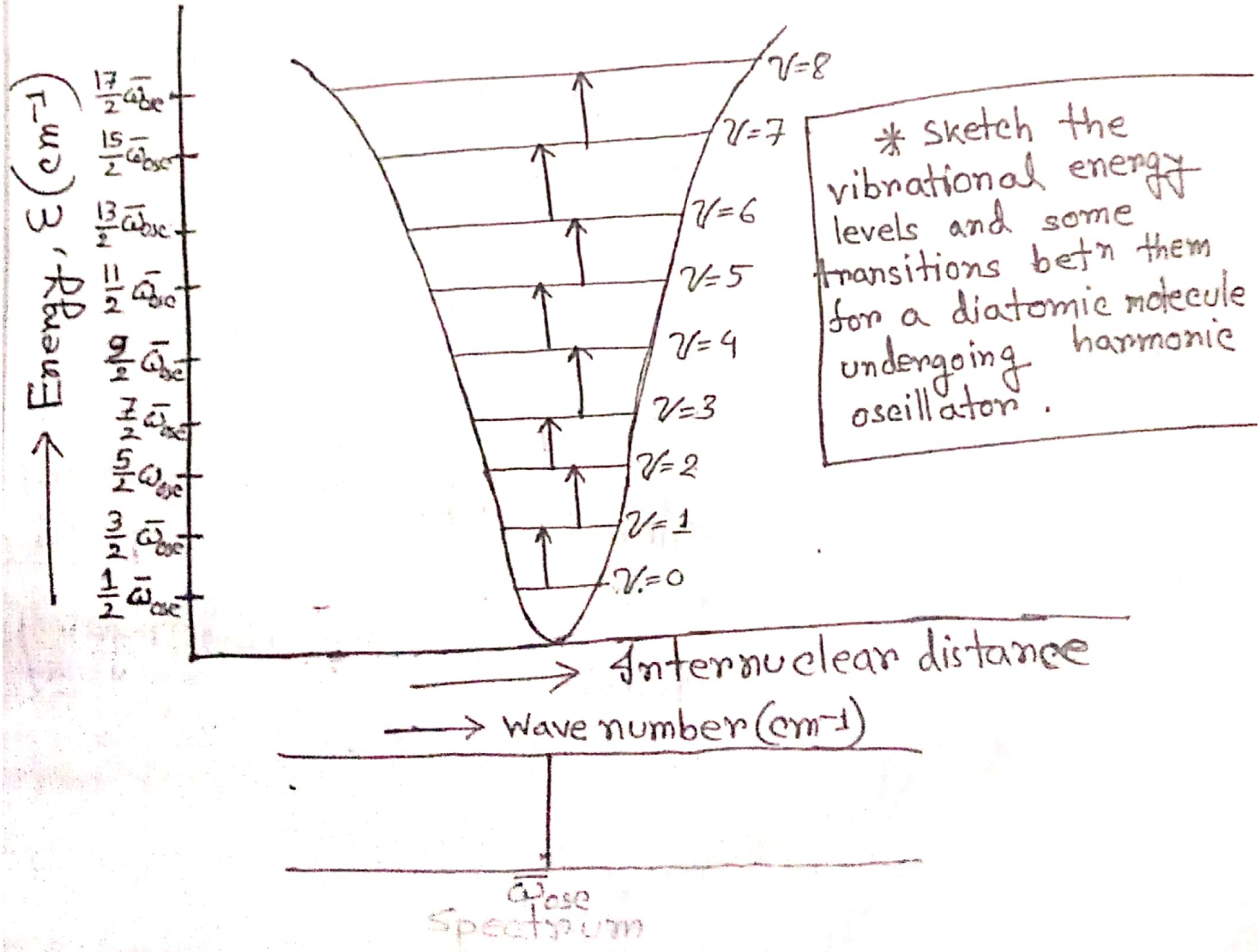
$$= 6.5 \times 10^{25} \text{ cm}^{-1}$$

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Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic and an anharmonic oscillators. Show the allowed transitions and the expected spectrum. (3)

Soln:

The schematic diagram for harmonic oscillator:



* Sketch the vibrational energy levels and some transitions betn them for a diatomic molecule undergoing harmonic oscillator.

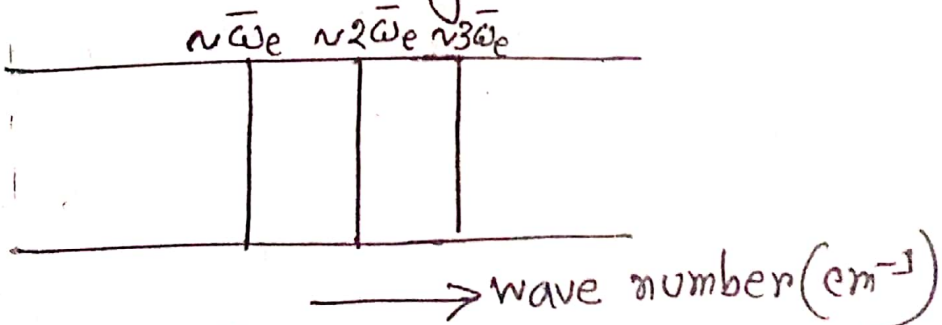
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Fig: The allowed vibrational energy levels and transition betn them for a diatomic molecule undergoing simple harmonic oscillation.

The schematic diagram for anharmonic oscillator:

The Morse curve

The spectrum arising from them,



Spectrum

Fig: The allowed vibrational energy levels and some transitions betn them for a diatomic molecule undergoing anharmonic oscillations.

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Explain why real molecules donot obey exactly the law of simple harmonic motion. (2)

Soln:

For real molecule vibration and rotation are not independent. Here two effects are appeared-

(1) Roto-vibrational coupling: When molecule vibrates, the bonds length are changing. Then the moment of inertia changes and the rotational constant B changes.

$$B_v = B_e - \alpha_e\left(v+\frac{1}{2}\right)$$

Where,

α_e = Roto-vibrational coupling constant.

B_e = Rotational constant for rigid and independent rotor.

Each vibrational level have different rotational constant, so $B_0 \neq B_1$. In this case, the energy of roto-vibrational level is,

$$E(v,J) = hc\left[\bar{\nu}_0\left(v+\frac{1}{2}\right) + B_e J(J+1) - \alpha_e\left(v+\frac{1}{2}\right) J(J+1)\right]$$

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② Centrifugal distortion:

When the radiation velocity increase (J increase) the bond length increase then the moment of inertia increase and the rotational constant B decrease.

$$B_J = B_e - DJ(J+1)$$

Here, D = Centrifugal distortion constant.

In this case, the energy of roto-vibrational level is,

$$E_{(v,J)} = h \cdot c \left[\bar{v}_0 \left(v + \frac{1}{2}\right) + B_e J(J+1) - DJ^2(J+1)^2 \right]$$

What are P, Q, R branches in the vibration-rotation spectra? Explain their origin. ②

Soln:

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P branch line:

lines to the low frequency side of band origin ($\bar{\omega}_0$), corresponding to negative m (that is $\Delta J = -1$) are reflected to as the P- branch.

Q- branch:

The lines arise for the transition, $\Delta J = 0$, is called the Q -branch.

R- branch:

lines to the high frequency side of band origin ($\bar{\omega}_0$), corresponding to positive m (that is $\Delta J = +1$) are referred to as the R branch.

P.T.O.

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lines arising from : $\Delta J = -2 \quad -1 \quad 0 \quad 1 \quad 2$

called : O P Q R S (branch)

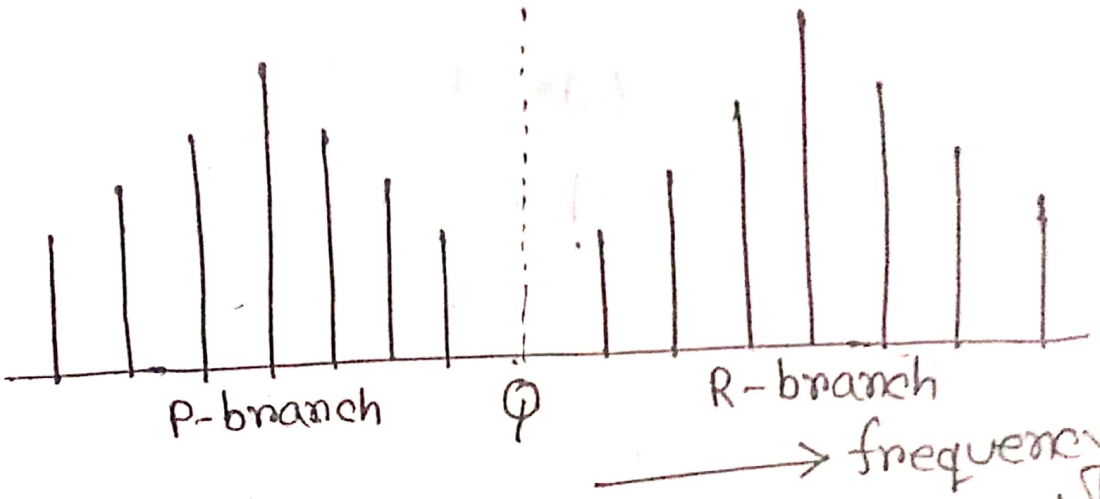


Figure : The formation of P, Q, R branch

Show the transitions for the P and R branches between the rotational-vibrational energy levels of a diatomic molecule and the spectrum arising from them. (2)

Sketch vibrational rotational energy levels and show the transitions between them to appear P and R branch lines in vibration-rotation spectrum. (2)

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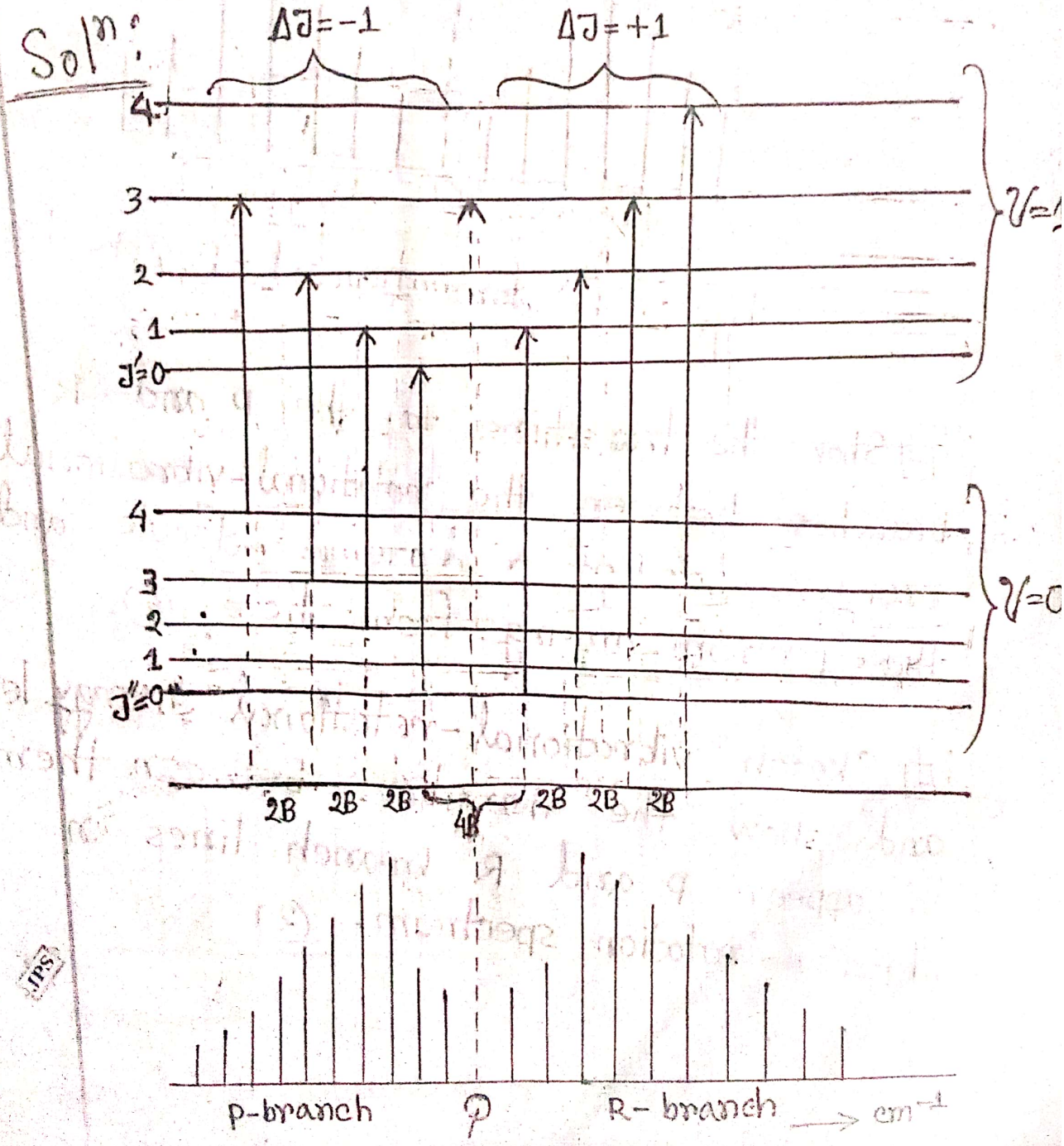
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Draw some transitions between the rotational-vibrational energy levels of a diatomic molecule together with the spectrum arising from them.

Soln:



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Fig: Transitions betn the rotational-vibrational energy levels of a diatomic molecule and the spectrum arising from them.

Explain the terms P and R branch line? When do they appear in the vibration-rotation spectrum? What is meant by band origin? (2+2+1) (2.5)

Soln:

P-branch line : (Previous)

R-branch line : (Previous)

line arising from : $\Delta J = -2\ -1\ 0\ 1\ 2$

called : O P Q R S

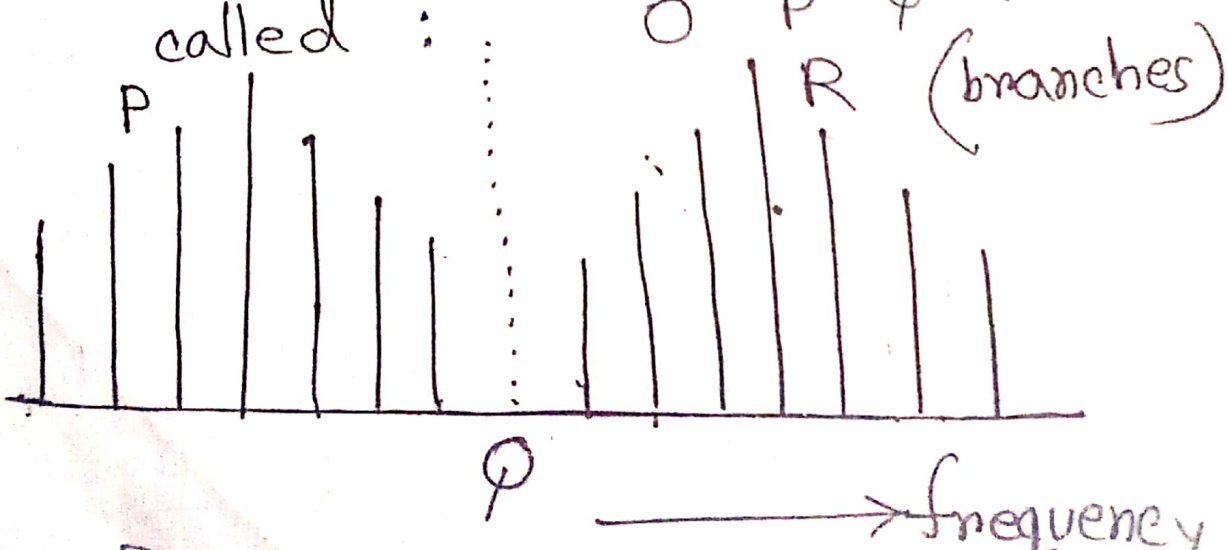


Fig: The formation of P, Q, R branch

They do appear in the vibration-rotation spectrum when—

P branch : $\Delta J = -1$, i.e. $J'' = J' + 1$ or $J' - J'' = 1$

$$\therefore \Delta E = \bar{\nu}_P = \bar{\omega}_0 - (B' + B'')(J'+1) + (B' - B'')(J'+1)^2 \quad cm^{-1}$$

where, $J' = 0, 1, 2 \ldots\ldots$

R branch : $\Delta J = +1$, $J' = J'' + 1$ or $J' - J'' = 1$

$$\therefore \Delta E = \bar{\nu}_R = \bar{\omega}_0 + (B' + B'')(J''+1) + (B' - B'')(J''+1)^2 \quad cm^{-1}$$

where, $J'' = 0, 1, 2 \ldots\ldots$

Here, $\bar{\nu}_P$ and $\bar{\nu}_R$ represents the wave number of P and R branch lines.

These two equation can be combined into-

$$\bar{\nu}_{P,R} = \bar{\omega}_0 + (B' + B'')m + (B' - B'')m^2 \quad cm^{-1}$$

where, $m = \pm 1, \pm 2 \ldots\ldots$

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Here, positive m values refer to the R branch and negative values to the P branch.

Band origin: The frequency $\bar{\omega}_0$ is usually called the band origin or band centre.

What is the basic requirement for a substance to be IR active? Discuss with example. (2) (1)

Soln:

Basic requirement for a substance to be IR active:

(i) The frequency of IR radiation must be equal to the vibrating frequency of a particular group.

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(ii) The molecule must have finite dipole moment.

(iii) If it doesnot have dipole moment, some of its vibrations must produce an induced dipole moment.

Examples:

Homonuclear diatomic molecules: N_2, O_2, H_2, F_2 double bond stretch in ethylene ($CH_2 = CH_2$) etc.

Explain why the C=O stretching vibration ~~of an aldehyde~~ of an aldehyde gives rise to a strong absorption in the infrared, although the absorption due to C=C vibration in an alkene is normally very week. (2) (1.5)

Soln:

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A signal is observed in the IR spectrum, if the dipole momentum of the molecule changes during the interaction with electromagnetic radiation. This is very likely for the groups which already posses a significant dipole momentum to start with e.g. C=O, C-Cl, O-H etc. Groups with a small or no difference in electronegativity e.g. C=C, C-H etc will usually show weak peaks in IR spectrum due to their less interaction with electromagnetic radiation.

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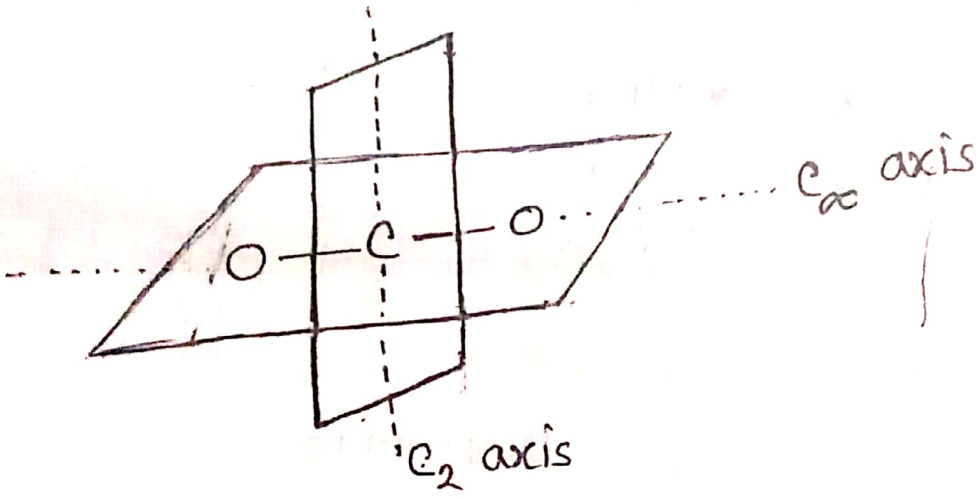
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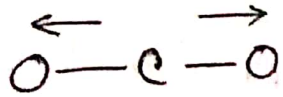
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Sketch the symmetry and fundamental vibrations of the carbon dioxide molecule in a figure. Explain them clearly. (2.25)

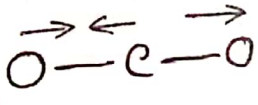
Soln:

Symmetry of CO_2 :

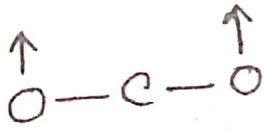




(a) Symmetric stretch
1330 cm^{-1}



(b) Antisymmetric stretch
2349.3 cm^{-1}



(c) Bending
667.3 cm^{-1}

Fig: The symmetry and fundamental vibration of the carbon dioxide molecule

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CO_2 molecule is a triatomic linear molecule. In this molecule, there are two different sets of symmetry axis. They are C_2 and C_∞ axis.

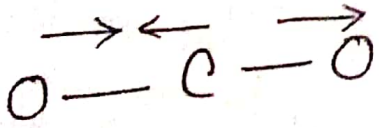
The molecule which have a centre of symmetry the dipole moment of them are zero. The symmetric bond stretching produce no change in dipole moment. So, this vibration is IR inactive and produce no IR spectrum. The antisymmetric bond stretching can change the dipole moment and produce IR spectrum.

The stretching vibration of CO_2 molecule is $N-1=2$ where, one is symmetric vibration and another is antisymmetric vibration. So we will get one IR spectrum for antisymmetric vibration because this vibration can change the dipole moment.

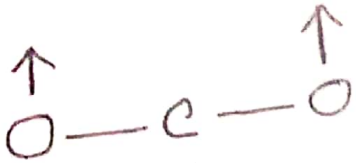
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The bending vibration of CO_2 molecule $= 2N-4 = 2$ and this two bending vibration are equivalent. One in the plane of the paper and other out of the plane. Dipole moment changes for bending vibration. So, we will get one IR spectrum from here.

So, total we get two IR band for CO_2 molecule instead of four.



Antisymmetric stretch.



Bending

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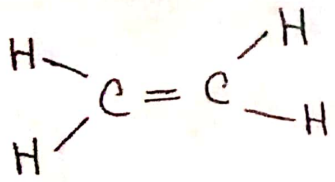
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[1] How many normal modes of vibration are possible for the following molecules:

$CH_2=CH_2$, NH_3 , OCS , C_6H_6 , SO_2 , HBr, O_2 , H_2O , BCl_3 , CH_3I . (2)

Soln:

(i) $CH_2=CH_2$



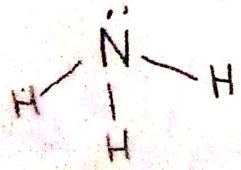
* A non-linear molecule of N-atoms have 3N-6 different internal vibrations.
* If the molecule is linear there are 3N-5 degrees of vibrational freedoms.

It is a non-linear molecule. So, the normal mode of vibration = 3N−6

$= (3 \times 6 - 6)$ [∵ N=6]

$= 12$

(ii) NH_3



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It is a non-linear molecule. So, the normal mode of vibration = 3N-6

$= (3 \times 4 - 6)$ [∵ N = 4]

$= 6$

(iii) OCS

$O = C = S$

It is a linear molecule. So, the normal mode vibration is = 3N-5

$= (3 \times 3 - 5)$ [∵ N=3]

$= 9 - 5$

$= 4$

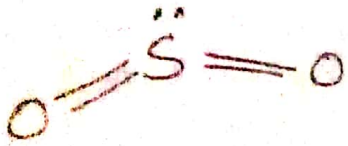
(iv) C_6H_6 ✓

It is non-linear molecule. So the normal mode of vibration is = 3N-6

$= 3 \times 12 - 6$ [∵ N=2]

$= 30$

(v) SO_2



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It is a non-linear molecule. So, the normal mode of vibration is = 3N-6

$= (3 \times 3 - 6)$ [$\because N = 3$]

$= 9 - 6$

$= 3$

(vi) HBr

It is a linear molecule. So, the normal mode of vibration is = 3N-5

$= 3 \times 2 - 5$ [$\because N = 2$]

$= 1$

(vii) H_2O

It is a non-linear molecule. So, the normal mode of vibration is = 3N-6

$= (3 \times 3 - 6)$ [$\because N = 3$]

$= 3$

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(viii) O_2

It is a linear molecule. So, the normal mode of vibration is $= 3N-5$

$= 3\times 2-5$ [$\because N=2$]

$= 1$

(ix) BCl_3.

It is a non-linear molecule. So, the normal mode of vibration $= 3N-6$

$=(3\times 4-6)$ [$\because N=4$]

$= 6$

(x) CH_3I

It is a non-linear molecule. So the normal mode of vibration $= 3N-6$

$=(3\times 5-6)$ [$\because N=5$]

$= 11$

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Find out the no. of fundamental vibration, stretching vibration and bending vibration of the following molecule–

(i) H_2O (ii) CO_2

Draw their fundamental vibrations and symmetry.

Soln:

(i) H_2O (Non-linear)

∴ Total number of fundamental vibration

$= 3N-6$

$= (3\times3-6)$ [∵N=3]

$= 3$

∴ The no. of stretching vibration $= N-1$

$= 3-1$ [∵N=3]

$= 2$

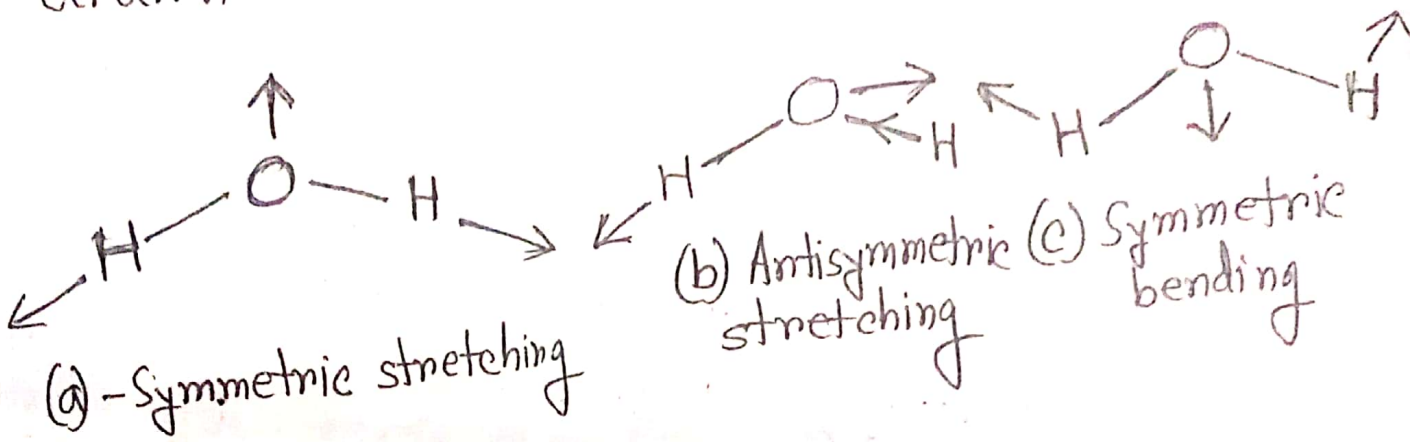
∴ The no. of bending vibration $= 2N-5$

$= (2\times3-5)$ [∵N=3]

$= 1$

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The fundamental vibrations of H_2O are drawn below :



(ii) CO_2 (linear)

Total no. of fundamental vibration

$= 3N-5$

$= 3 \times 3 - 5$

$= 4$

No. of stretching vibration $= N-1$

$= 3-1$

$= 2$

No. of bending vibration $= 2N-4$

$= 2 \cdot 3 - 4$

$= 2$

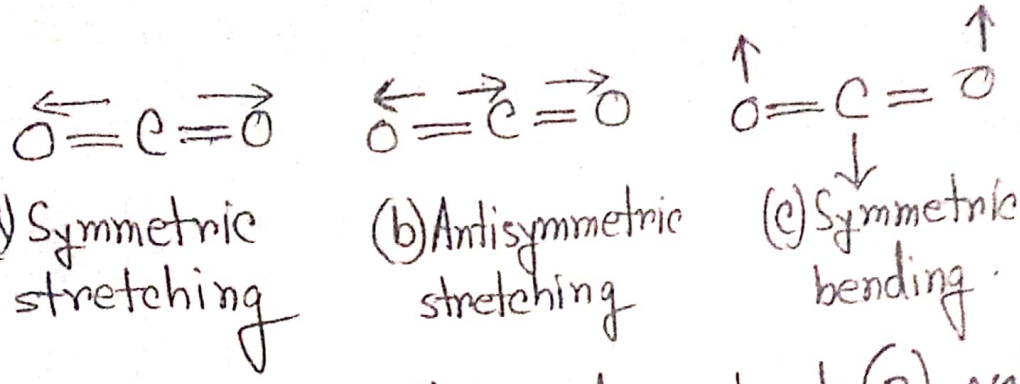
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The fundamental vibration of CO_2,



(a) Symmetric stretching (b) Antisymmetric stretching (c) Symmetric bending

HCl has rotational costant (B) value of 10.593 cm^{-1} and a centrifugal distortion constant D 5.3×10^{-4} cm^{-1} respectively. Estimate vibrational frequency of the molecule. (2)

Soln:

We know, centrifugal distortion, $D = \frac{4B^3}{\bar{\omega}_e^2}$

$\therefore$ Vibrational frequency, $\bar{\omega}_e = \sqrt{\frac{4B^3}{D}}$

$\therefore \bar{\omega}_e = \sqrt{\frac{4 \times (10.593)^3}{5.3 \times 10^{-4}}} = 2995.2\ cm^{-1}$

$= 2995.2 \times 10^2\ m^{-1}$

Extra:

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m_H m_{Cl}}{m_H + m_{Cl}} = \frac{1.673 \times 10^{-27} \times 58.06 \times 10^{-27}}{1.673 \times 10^{-27} + 58.06 \times 10^{-27}}$$

$= 1.6261 \times 10^{-27}$

$\therefore$ Force constant, $k = 4\pi^2 \bar{\omega}_e^2 c^2 \mu$

$= 4\pi^2 \times (2995.2 \times 10^2)^2 \times (3 \times 10^8)^2 \times (1.6261 \times 10^{-27})$

$= 518.3\ Nm^{-1}$

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- How vibrational spectrum of a diatomic molecule enables us to determine anharmonicity constant and equilibrium frequency of vibration? (3)

[अथवा]

- Explain why all transitions in vibrational spectroscopy are restricted to originate from ground state ($v=0$). (2)

[अथवा]

- How can the intermolecular and intramolecular hydrogen bonding be identified by IR spectroscopy.

[अथवा].

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